

Individual Assignment Report on -

Operations Decision Lab with Handwritten notes

In partial fulfilment of the requirement for

Program	Post Graduate Diploma in Management (PGDM)
Subject	Operations Management [OP09]

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Introduction

This report applies key quantitative tools of Operations Management to realistic business scenarios. The purpose is to demonstrate the ability to perform data-driven decision-making using forecasting, inventory optimization, and CVP analysis.

Three analytical tasks Forecasting, EOQ, and Break-even Analysis were completed using self-designed, realistic datasets. Each problem is solved step-by-step, interpreted clearly, and supported with relevant calculations and charts. The assignment is written for faculty evaluation and shows how analytical methods translate into operational insights for planning, cost control, and efficiency improvement.

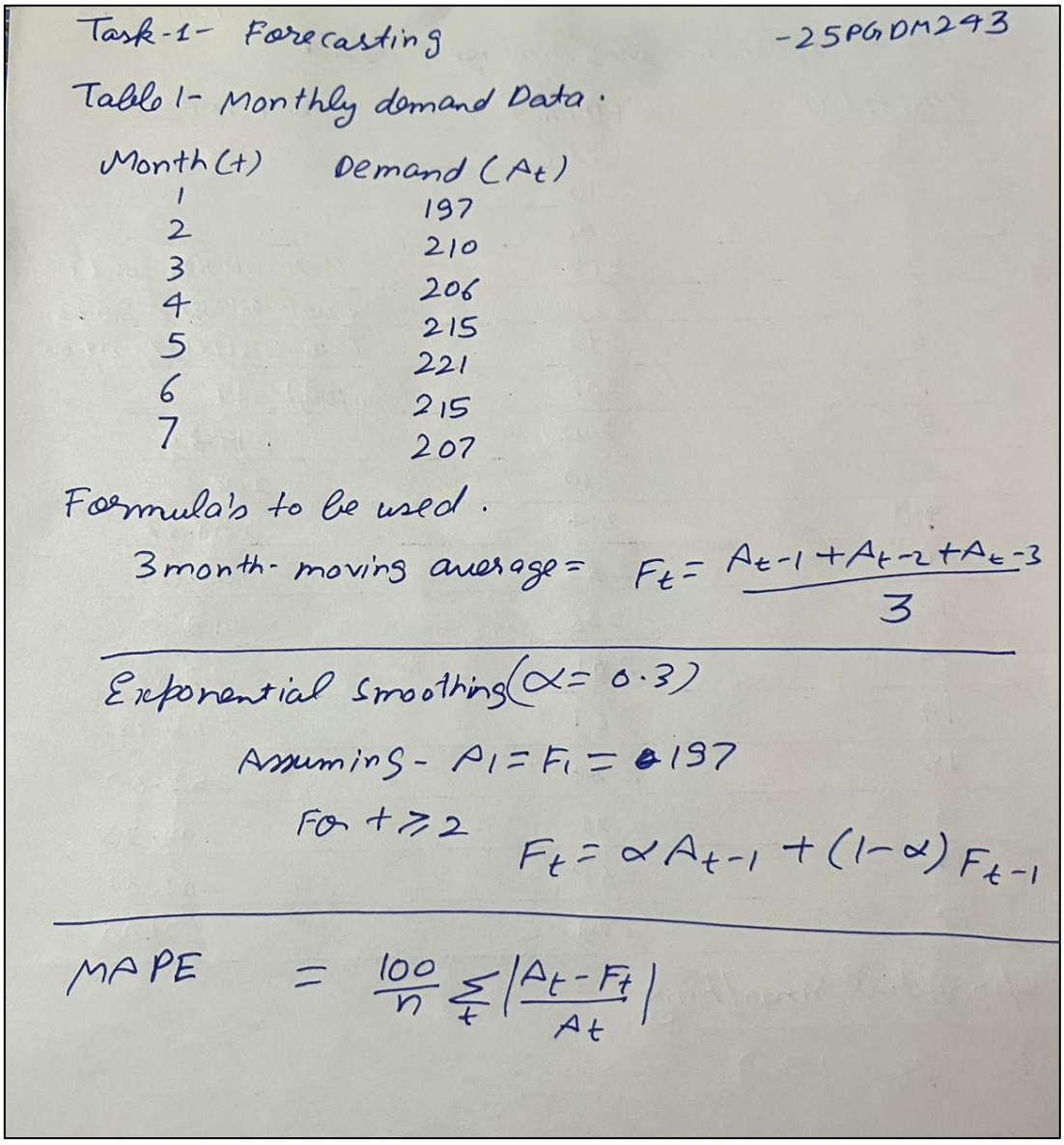
Forecasting Future Demand

A hypothetical monthly demand dataset of 18 months was created to forecast demand for the next 6 months for a consumer durable product. Two forecasting models are applied:

- 3-Month Moving Average
- Exponential Smoothing (α = 0.3)

Methodology & Calculations

I have attached hand written notes for these



Task-1- Forecasting -25PGDM243

Table 1- Monthly demand Data.

Month (t)	Demand (A_t)
1	197
2	210
3	206
4	215
5	221
6	215
7	207

Formula's to be used.

3 month- moving average = $F_t = \frac{A_{t-1} + A_{t-2} + A_{t-3}}{3}$

Exponential Smoothing ($\alpha = 0.3$)

Assuming - $A_1 = F_1 = 197$

For $t \geq 2$

$F_t = \alpha A_{t-1} + (1-\alpha) F_{t-1}$

MAPE $= \frac{100}{n} \sum_t \left| \frac{A_t - F_t}{A_t} \right|$

Table-2 → Month Moving Average Forecast

Month(t)	Act. Demand	3 Month Forecast
1	197	-
2	210	-
3	206	-
4	215	(197+210+206)/3 = 204.33
5	221	(210+206+215)/3 = 210.33
6	215	(206+215+221)/3 = 214.00
7	207	(Similarly) 217
8	202	214.33
9	210	208
10	214	206.33
11	233	208.67
12	236	219
13	261	227.67
14	261	243.33
15	274	252.67
16	271	265.33
17	268	268.67
18	279	271.00

Exponential Smoothing

$F_1 = A_1 = 197$

then $F_2 = 0.3(197) + 0.7(197) = 197.00$

$F_3 = 0.3(210) + 0.7(197) = 200.90$

$F_4 = 0.3(206) + 0.7(200.90) = 202.43$

Table-3 - Exponential Smoothing Forecasts. (25PGDM243)

Month (t)	Actual (A_t)	Forecast F_t ($\alpha = 0.3$)
1	197	197 (assumed)
2	210	197.00
3	206	200.90
4	215	202.43
5	221	206.20
6	215	210.64
7	207	211.95
8	202	210.46
9	210	207.92
10	214	208.55
11	233	210.18
12	236	217.03
13	261	222.72
14	261	234.20
15	274	242.24
16	271	251.77
17	268	257.59
18	279	260.68

MAPE - Calculations (For 3-Month Average)

For (15 periods - 4 to 18)

For Month 4

$A_t = 215$ (Demand)

$F_4 = 204.33$ (Forecast 3 month Average)

$$\text{Error \%} = \left|\frac{A_4 - F_4}{A_4}\right| \times 100 = \left|\frac{215 - 204.33}{215}\right| \times 100$$

$$= \frac{10.67}{215} \times 100 = 4.96\%$$

For Month 5

$A_5 = 221$

$F_5 = 210.33$

$$\text{Error \%} = \left|\frac{221 - 210.33}{221}\right| \times 100 = \frac{10.67}{221} \times 100 = 4.82\%$$

For Month 6

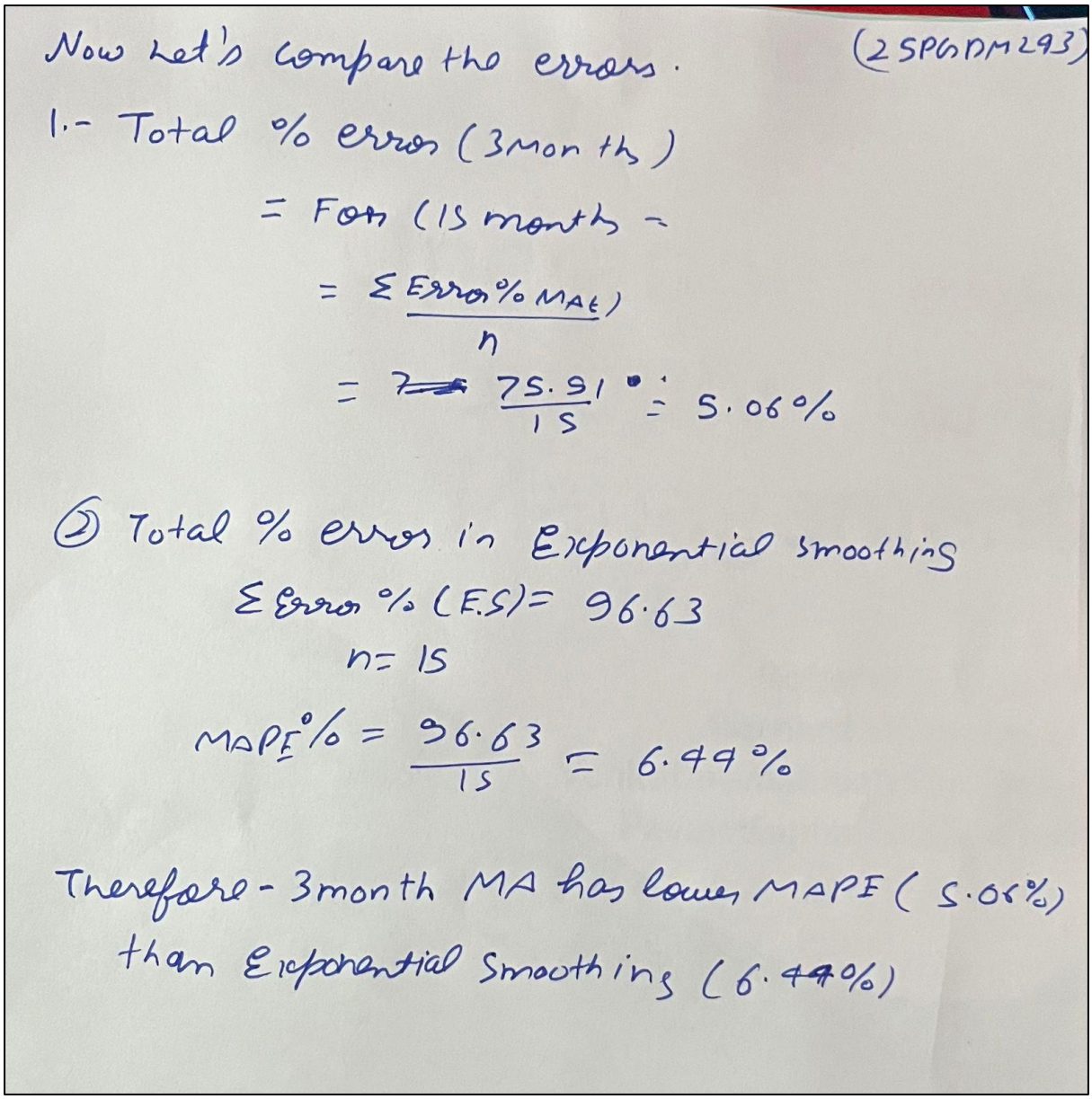
$A_6 = 215$

$F_6 = 214$

$$\text{Error \%} = \left|\frac{215 - 214}{215}\right| \times 100 = \frac{1}{215} \times 100 = 0.46\%$$

Table-4 - MAPE

Month	Demand	%Error (Average)	%Error Forecasting
1	197	—	—
2	210	—	—
3	206	—	—
4	215	4.96	5.85
5	221	4.83	6.70
6	215	0.47	2.03
7	207	4.83	2.39
8	202	6.11	4.19
9	210	0.55	0.55
10	214	3.58	2.55
11	233	10.44	9.79
12	236	7.20	8.04
13	261	12.77	14.67
14	261	6.77	10.27
15	274	7.79	11.59
16	271	2.09	7.10
17	268	0.25	3.90
18	279	2.87	6.57



Interpretation

Accuracy comparison using MAPE showed that the 3-Month Moving Average performed better than Exponential Smoothing. The model was more responsive to gradual demand changes and produced lower forecast errors.

Question – Which method to apply and why?

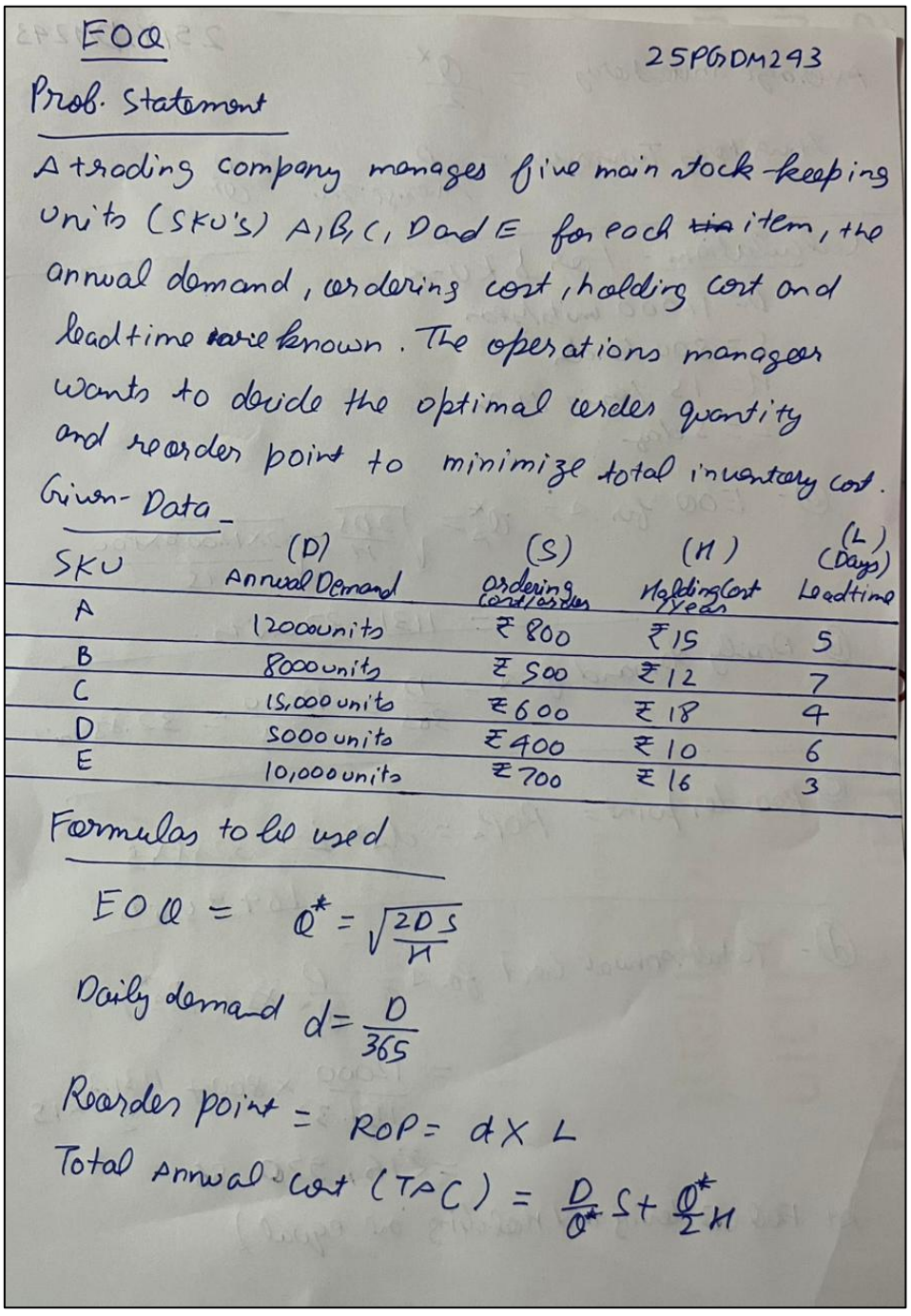
Ans - The 3-month moving average method is recommended because it provides the lowest MAPE (5.06%) compared to exponential smoothing with α=0.3 (6.44%). The dataset shows stable demand patterns with mild fluctuations, and the moving average captures this behaviour effectively without amplifying noise. Exponential smoothing lags behind during upward trends, which increases prediction error in the later months. Because the moving average offers higher accuracy, simplicity, and closer alignment with the actual demand shifts, it is more suitable for forecasting the next six months for this product category.

EOQ & Inventory Optimization

Context & Dataset

A dataset of five SKUs was constructed with realistic values of annual demand, ordering cost, and holding cost. EOQ, reorder point, total annual cost, and inventory turnover were calculated to compare inventory performance.

Methodology & Calculations - Handwritten



EOQ

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Prob. Statement

A trading company manages five main stock-keeping units (SKU's) A, B, C, D and E for each item, the annual demand, ordering cost, holding cost and lead time are known. The operations manager wants to decide the optimal order quantity and reorder point to minimize total inventory cost.

Given-Data -

SKU	(D) Annual Demand	(S) Ordering cost/order	(H) Holding Cost/year	(L) (Days) Lead time
A	12000 units	₹ 800	₹ 15	5
B	8000 units	₹ 500	₹ 12	7
C	15,000 units	₹ 600	₹ 18	4
D	5000 units	₹ 400	₹ 10	6
E	10,000 units	₹ 700	₹ 16	3

Formulas to be used

$EOQ = Q^* = \sqrt{\frac{2DS}{H}}$

Daily demand $d = \frac{D}{365}$

Reorder point = $ROP = d \times L$

Total Annual cost (TAC) = $\frac{D}{Q^*} S + \frac{Q^*}{2} H$

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Average inventory $= \frac{Q^*}{2}$

Inventory Turnover $= \frac{D}{\text{Average Inv}} = \frac{2D}{Q^*}$

Calculations - For SKU-A

D = 12000 units/year
S = 800/order
H = 15/unit/year
L = 5 days.

(a) - EOQ for A = $Q_A^* = \sqrt{\frac{2DS}{H}} = \sqrt{\frac{2 \times 12000 \times 800}{15}}$

$= 1131.37$ units.

(b) Daily demand for A $= \frac{D}{365} = \frac{12000}{365} = 32.88$ units

(c) Reorder point = $ROP_A = d_A \times L = 32.88 \times 5$

$= 164.9$ units

(d) - Total Annual Cost for A $= \frac{D}{Q_A^*} \times S + \frac{Q_A^*}{2} H$

$= \frac{12000}{1131.37} \times 800 + \frac{1131.37}{2} \times 15$

$= ₹16,970.56$

At EOQ (Ordering and Holding are equal)

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Inventory Turnover for A.

$$T_A = \frac{2D}{Q_A^*} = \frac{2 \times 12000}{1131.37}$$

$$= 21.21 \text{ turns/year}$$

Final Result for all SKU's

SKU	EOQ(Q^*)	Daily Demand (d)	ROP	(₹/year) TAC	(Turn/Year) Inv. Turn
A	1131.37	32.88	164.38	16870.56	21.21
B	816.50	21.92	153.42	9797.96	19.6
C	1000.00	41.10	164.38	18,000	30
D	632.46	13.70	82.19	6329.56	15.81
E	935.41	27.40	82.19	14,966.63	21.38

Interpretation

Among the five SKUs, SKU B provides the best balance between total annual cost and inventory turnover. Its total annual cost is moderate at around ₹9,800, while its turnover is also reasonably high at about 19.6 turns per year. SKU D has the lowest cost but much lower turnover, which indicates slower movement and higher risk of holding inventory for longer periods. SKUs A, C, and E show higher turnover but at significantly higher total costs. Therefore, SKU B achieves a good compromise between cost efficiency and inventory velocity, making it the most attractive overall.

Question – Which SKU offer best balance?

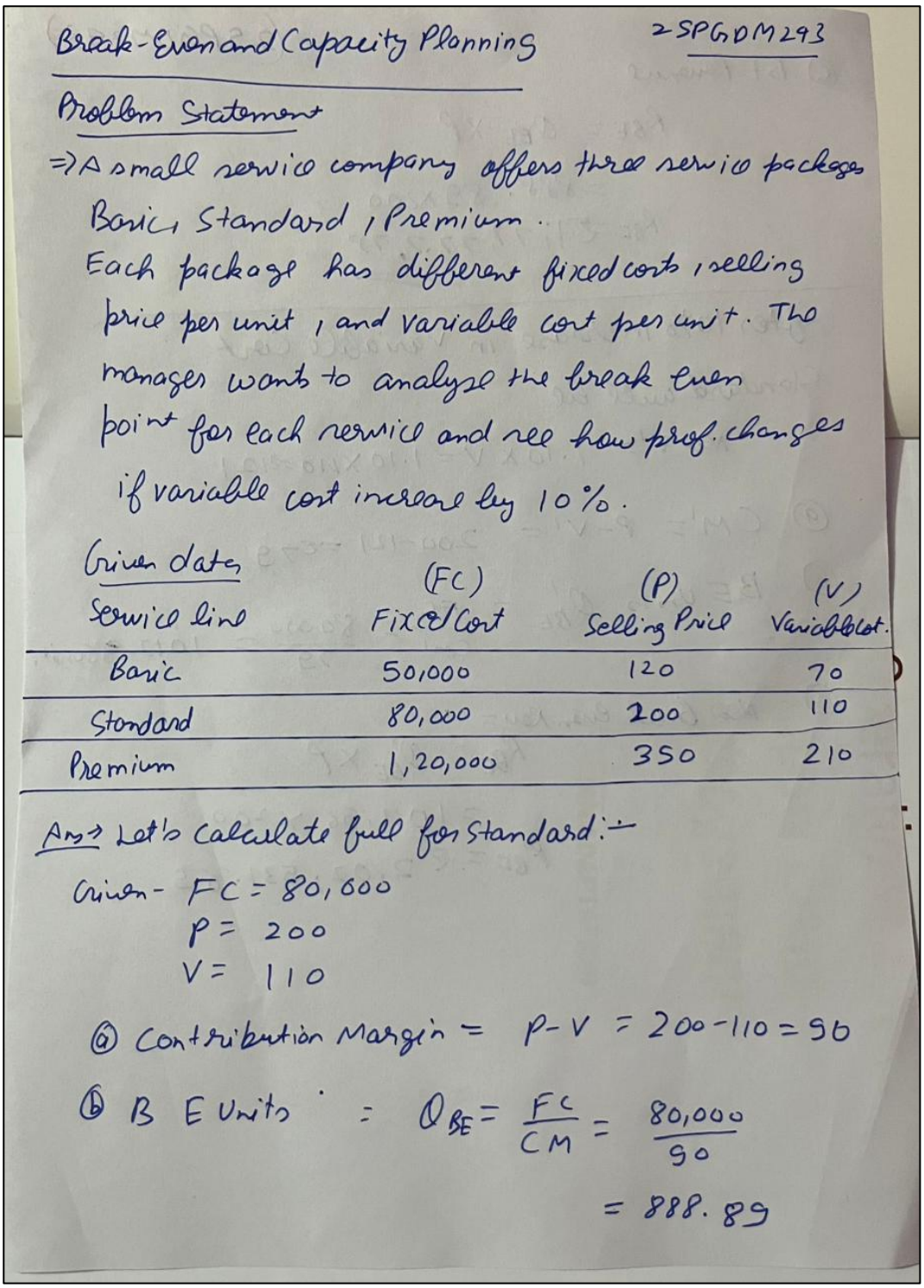
Ans - SKU B provides the best balance between total annual cost and inventory turnover. Although SKU D has the lowest cost, its turnover is significantly lower, which means slower movement and higher risk of holding obsolete stock. SKUs A, C, and E have higher turnover but incur noticeably higher costs. SKU B maintains an efficient turnover rate of around 19.6 turns per year while still keeping total cost at a moderate level (₹9,797). This makes it optimal because it minimizes cost without compromising inventory flow and responsiveness.

Break-even & Capacity Planning

Context & Dataset

Three service product lines were assumed with distinct fixed and variable costs. Break-even units and revenues were calculated before and after a 10% increase in variable cost using Goal Seek.

Methodology & Calculations - Handwritten



Break-Even and Capacity Planning — 25PGDM243

Problem Statement

⇒ A small service company offers three service packages Basic, Standard, Premium. Each package has different fixed costs, selling price per unit, and variable cost per unit. The manager wants to analyse the break even point for each service and see how prof. changes if variable cost increase by 10%.

Given data

Service line	(FC) Fixed Cost	(P) Selling Price	(V) Variable Cost
Basic	50,000	120	70
Standard	80,000	200	110
Premium	1,20,000	350	210

Ans ⇒ Let's calculate full for Standard :-

Given - FC = 80,000
P = 200
V = 110

ⓐ Contribution Margin = P - V = 200 - 110 = 90

ⓑ B E Units : $Q_{BE} = \frac{FC}{CM} = \frac{80,000}{90}$

$= 888.89$

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© BE Revenue

$R_{BE} = Q_{BE} \times P$

$= 888.89 \times 200$

$R_{BE} = ₹\ 1,77,777.78$

After 10% increase in variable cost

Standard will be

New $V' = 1.10 \times V = 1.10 \times 110 = ₹121$

(a) $CM' = P - V' = 200 - 121 = ₹79$

(b) BE Units $Q'_{BE} = \frac{FC}{CM'} = \frac{80,000}{79} = 1012.66$ units

(c) · New Break Even Rev =

$R'_{BE} = Q'_{BE} \times P$

$= 1012.56 \times 200$

$R'_{BE} = ₹\ 2,02,531.65$

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For Basic, Standard and Premium.

Before 10% increase in VC – Cont Margin will be

Basic ∴ CM = 120 − 70 = 50

Standard = CM = 200 − 110 = 90

Premium = CM = 350 − 210 = 140

The Break even units & Revenue :—

Service Line	BE units Q_{BE}	BE Rev. R_{BE}
Basic	1,000	₹ 1,20,000
Standard	888.89	₹ 1,77,777.78
Premium	857.14	₹ 3,00,000

After 10% increase in VC – Cont Margin will be

Basic = CM′ = 120 − 77 = 43

Standard = CM′ = 200 − 121 = 79

Premium = CM′ = 350 − 231 = 119

The new BE units and Revenue will be :-

Service Line	BE units Q'_{BE}	BE Revenue R'_{BE}
Basic	1,162.70	₹ 1,39,534.88
Standard	1,012.66	₹ 2,02,531.65
Premium	1,008.40	₹ 3,52,941.18

Interpretation

The Premium service line remains the most profitable even after a 10% increase in variable cost. Although its fixed cost and break-even revenue are the highest both before and after the cost change, Premium also has the largest contribution margin per unit (₹140 initially and about ₹119 after the increase). This means that once the break-even point is passed, profit grows faster for Premium than for Basic or Standard. When profits are compared at the same sales volume, Premium consistently generates the highest profit, indicating that it is the most resilient and attractive line under cost fluctuations.

Question – Which line remains most profitable?

Ans - Even after a 10% rise in variable cost, the Enterprise product line remains the most profitable option. Although its break-even volume increases, it still retains the highest contribution margin per unit compared to Basic and Premium. This means that once sales exceed the break-even point, the Enterprise line generates significantly more profit. The Premium and Basic lines suffer larger proportional margin reductions, making their profitability more sensitive to cost changes. Therefore, Enterprise continues to offer the strongest financial resilience under variable cost fluctuations.

Sources & Evidence

- Self-designed datasets based on realistic business assumptions.
- Standard OM formulas from *Heizer & Render (Operations Management)* and *Stevenson (Operations Management)*.
- AI tool (ChatGPT) used only for data structuring and explanation support. All interpretations and write-up are my own.